

# MATH 112B

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## 1

### 1.1 Lecture 1 - Gradient Systems

Function of several variables:

### 1.2 Lecture 2 - Introduction to PDEs

#### Conservation Laws

This is used to describe: bacterial movement in a petridish, nerve signal traveling through an axon, heat diffusion in a rod, cars on the highway.

We start with the following assumptions:

- Motion happens in 1D.
- Cross-sectional area is constant.

Consider a 1D tube, with particles inside. How do we describe the change of concentration of particles?

$$\begin{array}{c} \text{change in pop between} \\ x \text{ and } x + \Delta x \end{array} = \begin{array}{c} \text{rate of entry into} \\ (x, x + \Delta x) \\ \text{per unit time} \end{array} - \begin{array}{c} \text{rate of exit into} \\ (x, x + \Delta x) \\ \text{per unit time} \end{array}$$

Define

- $c(x, t)$  = concentration of particles at  $x$  at time  $t$
- $J(x, t)$  = flux of particles at  $(x, t)$  = # of particles crossing a unit area at  $x$  in the positive direction, per unit time.
- $A$  = cross-sectional area of the tube.
- $\Delta V$  = volume of our segment =  $A \cdot \Delta x$

We have

$$\begin{aligned} \frac{\partial}{\partial t} \underbrace{(c(x, t) \cdot A \cdot \Delta x)}_{\text{\# of particles}} &= J(x, t)A - J(x + \Delta x, t)A \\ \frac{\partial c}{\partial t} &= \frac{J(x, t) - J(x + \Delta x, t)}{\Delta x} \\ \lim_{\Delta x \rightarrow 0} \frac{\partial c}{\partial t} &= \lim_{\Delta x \rightarrow 0} \frac{J(x, t) - J(x + \Delta x, t)}{\Delta x} \\ \frac{\partial c}{\partial t} &= -\frac{\partial J}{\partial x} \leftarrow \text{Conservation of mass} \end{aligned}$$

Consider a new term,

$$\begin{array}{c} \Delta \text{pop between} \\ x \text{ and } x + \Delta x \end{array} = \begin{array}{c} \text{rate of entry into} \\ (x, x + \Delta x) \\ \text{per unit time} \end{array} - \begin{array}{c} \text{rate of exit into} \\ (x, x + \Delta x) \\ \text{per unit time} \end{array} \pm \begin{array}{c} \text{rate of local} \\ \text{creation} \\ \text{or degradation} \end{array}$$

We now have

- $\sigma(x, t)$  = # of particles created/removed per unit volume at time  $t$

Now our conservation laws become

$$\frac{\partial c}{\partial t} = -\frac{\partial J}{\partial x} \pm \sigma(x, t)$$

To generalize to 2D or 3D:

$$\frac{\partial c}{\partial t} = -\vec{\nabla} \cdot \vec{J} \pm \sigma(\vec{x}, t)$$

Now we have  $c = c(\vec{x}, t)$

$$\sigma = \sigma(\vec{x}, t)$$

$$\vec{J} = (J_x, J_y, J_z)$$

$$\vec{\nabla} = \left( \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right)$$

So we have

$$\frac{\partial c}{\partial t} = -\left( \frac{\partial J_x}{\partial x} + \frac{\partial J_y}{\partial y} + \frac{\partial J_z}{\partial z} \right) \pm \sigma(x, y, z, t)$$

ex Consider a moving fluid + organisms that move with the flow (not actively). Suppose that the velocity of the fluid is given by

$$\vec{v}(x, y, z, t)$$

Let  $c(x, y, z, t)$  = concentration of organism.

What is the flux?

$$\begin{aligned} \vec{J} &= c \cdot \vec{V} \quad \text{Assume } \sigma = 0 \\ \Rightarrow \frac{\partial c}{\partial t} &= -\left( \frac{\partial c V_x}{\partial x} + \frac{\partial c V_y}{\partial y} + \frac{\partial c V_z}{\partial z} \right) \end{aligned}$$

In 1D:

$$\begin{aligned} \frac{\partial c}{\partial t} &= -\frac{\partial}{\partial x}(cV) \\ &= -\frac{\partial c}{\partial x}V - \frac{\partial V}{\partial x}c \end{aligned}$$

What kind of PDE is this?

- 1st order
- 2 variables:  $x, t$
- Linear

- Transport equation

[ex] Attraction/repulsion. Suppose that  $\psi$  is a source of attraction for the bacteria. The motion is determined by the gradient of  $\psi$ .

In 1D:  $\psi = \psi(x, t)$ ,  $J = \frac{\partial \psi}{\partial x}$

$$J = c\alpha \vec{\nabla} \psi$$

We have

$$\begin{aligned} \frac{\partial c}{\partial t} &= -\frac{\partial J}{\partial x} \\ \frac{\partial c}{\partial t} &= -\frac{\partial}{\partial t} \left( c\alpha \frac{\partial \psi}{\partial x} \right) \end{aligned}$$

$\psi$  is given.

$$\frac{\partial c}{\partial t} = -\alpha \frac{\partial c}{\partial x} \cdot \frac{\partial \psi}{\partial x} - \alpha c \frac{\partial c}{\partial x} \cdot \frac{\partial^2 \psi}{\partial x^2}$$

This is 1st order, linear PDE in 2 variables.

[ex] Temperature of a heated rod.

Here we define  $c(x, t)$  as concentration of energy/temperature. And since heat spreads from hot to cold, we get the following.

Fick's Law :  $J = -D \cdot \frac{\partial c}{\partial x}$

Then we have

$$\begin{aligned} \frac{\partial c}{\partial t} &= -\frac{\partial}{\partial x} \left( -D \frac{\partial c}{\partial x} \right) \\ &= D \frac{\partial^2 c}{\partial x^2} \leftarrow \text{Heat equation} \end{aligned}$$

In higher dimensions:

$$\begin{aligned} \frac{\partial c}{\partial t} &= \vec{\nabla} (D \vec{\nabla} c) \\ &= D \left( \frac{\partial^2 c}{\partial x^2} + \frac{\partial^2 c}{\partial y^2} + \frac{\partial^2 c}{\partial z^2} \right) \end{aligned}$$

[ex] Random motion of particles.

Denote  $\lambda_r$  as the amount of particles jumping from  $x \rightarrow x + \Delta x$

Denote  $\lambda_l$  as the amount of particles jumping from  $x \rightarrow x - \Delta x$

Denote  $c(x, t) = \#$  of particles within  $(x, x + \Delta x)$  at time  $t$ :

$$c(x, t + \Delta t) = c(x, t) + \lambda_r \cdot c(x - \Delta x, t) + \lambda_l \cdot c(x + \Delta x, t) - \lambda_l c(x, t) - \lambda_r c(x, t)$$

Using Taylor Series:

$$\begin{aligned} c(x, t + \Delta t) &= c(x, t) + \frac{\partial c}{\partial t} \Delta t + \frac{1}{2} \frac{\partial^2 c}{\partial t^2} (\Delta t)^2 + \dots \\ c(x + \Delta x, t) &= c(x, t) + \frac{\partial c}{\partial x} \Delta x + \frac{1}{2} \frac{\partial^2 c}{\partial x^2} (\Delta x)^2 + \dots \\ c(x - \Delta x, t) &= c(x, t) - \frac{\partial c}{\partial x} \Delta x + \frac{1}{2} \frac{\partial^2 c}{\partial x^2} (\Delta x)^2 + \dots \end{aligned}$$

We can simplify using  $\lambda_r = \lambda_l = \lambda$ , then we have

$$\begin{aligned} c + \frac{\partial c}{\partial t} \Delta t + \frac{\partial^2 c}{\partial t^2} \frac{(\Delta t)^2}{2} + \dots &= c \\ &+ \lambda \left( c + \frac{\partial c}{\partial x} \Delta x + \frac{1}{2} \frac{\partial^2 c}{\partial x^2} (\Delta x)^2 \right) \\ &+ \lambda \left( c - \frac{\partial c}{\partial x} \Delta x + \frac{1}{2} \frac{\partial^2 c}{\partial x^2} (\Delta x)^2 \right) \\ &- \lambda c - \lambda c \\ \frac{\partial c}{\partial t} \Delta t + \frac{\partial^2 c}{\partial t^2} \frac{(\Delta t)^2}{2} &= \lambda \frac{\partial^2 c}{\partial x^2} (\Delta x)^2 \end{aligned}$$

Call  $\lim_{\Delta x \rightarrow 0, \Delta t \rightarrow 0} \frac{(\Delta x)^2}{\Delta t} \equiv K$  and ignore second order time term as we take the limit.

$$\begin{aligned} \frac{\partial c}{\partial t} \Delta t &= \lambda (\Delta x)^2 \cdot \frac{\partial^2 c}{\partial x^2} \\ \frac{\partial c}{\partial t} &= \lambda K \cdot \frac{\partial^2 c}{\partial x^2} \leftarrow \text{Diffusion equation} \end{aligned}$$